TERM PROJECT REPORT: Hybrid Bomb Defusal Simulation

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1. Project Overview and Motivation

1.1. What is the Project?

Our project, "Defuse the Bomb," is a hybrid simulation game that combines physical hardware interaction with digital software puzzles. The player acts as a bomb disposal expert who must multitask between solving logic puzzles on a computer screen and physically manipulating a bomb circuit (cutting wires, pressing buttons) before the countdown timer reaches zero.

1.2. The Need (Why did we do this?)

In engineering, software and hardware are often taught separately. We wanted to create a project that bridges this gap. The main goals of this project were:

- To learn how to establish real-time communication between a PC (Python) and Microcontrollers (Arduino).
- To understand sensor integration (MPU6050) and circuit logic.
- To practice multitasking in game design.
- To experience working as a large team, dividing tasks into hardware, software, and documentation roles.

2. System Architecture

The system operates on a "PC-Arduino" architecture:
- PC/Python: Runs the main game interface, logic puzzles, and processes sensor data. It acts as the brain.
- Arduinos: There are two separate Arduino units acting as bodies. They handle physical inputs/outputs and send sensor data to the PC.

2.1. Hardware Design (The Solution)

We encountered a technical limitation during the design phase. The MPU6050 Tilt Sensor requires specific pins for communication. However, our main bomb circuit already used those pins for the penalty button and alarm system. Also, the MPU6050 game required a higher speed for smooth control.

Our Solution: We decided to use a Dual-Arduino Setup:

1. **Main Control Unit (Arduino Uno 1):**
   Controls the TM1637 countdown display, manages the 5-wire defusal mechanism, runs the "Simon Says" memory game using 4 LEDs and 4 Buttons, and communicates via Serial at 9600 Baud.
2. **Sensor Unit (Arduino Uno 2):**
   Dedicated solely to the MPU6050 Accelerometer. Reads tilt data (X/Y axis) and sends it to the PC at a high speed (115200 Baud) for smooth control in the Maze Game.

2.2. Software Design

The PC software is written in Python. It uses a multi-threaded approach to listen to two different USB ports (COM Ports) simultaneously.
Libraries Used: tkinter (GUI), pygame (Game Physics), pyserial (Communication), opencv (Camera/Motion Detection).

3. How It Works (The Gameplay Flow)

The game consists of 4 digital levels. After completing each level on the screen, the system unlocks a physical wire on the Arduino bomb that must be cut.

3. Start: The player presses the START button on the PC. The Python script sends an 'S' command to the Arduino. The Arduino starts the 5-minute countdown.
4. Level 1 - Laser Mirrors: A logic puzzle where the player reflects a laser beam.
5. Level 2 - Tilt Arena: The player controls a ball by physically tilting the sensor box.
6. Level 3 - Energy Stabilizer: A resource management game.
7. Level 4 - Red Light Green Light: Uses the laptop camera to detect player motion.
8. Final Stage - Simon Says: Once all PC tasks are done, the Arduino enters 'Simon Mode'. The player must repeat blinking LED sequences.

4. Communication Protocol

We created a custom Serial protocol to synchronize the PC and Arduino.

From Python to Arduino:

- 'S': Start timer | '0'-'3': Set Target Wire | 'W': Activate Simon | 'P': Panic/Explode

From Arduino to Python:

- TIME: [seconds]: Syncs PC timer | WIRE_CORRECT: Unlocks next level | BOOM!: Game Over | YOU WON!: Success

5. Group Members and Responsibilities

Our group worked collaboratively, attending regular meetings to integrate the parts.

- **Hardware & Circuit Design Team (Ahmet Burak Sağlam, Begüm Çakmak, Volkan Baş):**
  Responsible for the Main Control Unit (Bomb) architecture. Designed the circuit, assembled the components, and developed the firmware which controls the countdown timer, wire cutting logic, and the 'Simon Says' hardware module. Managed complex cable connections and hardware stability. Assembled all the parts into a cardboard bomb model.

- **Software Architecture & Integration (Arda Ay):**
  Developed the main Python Interface (GUI), implemented multi-threading architecture to allow dual-Arduino communication, and integrated the data streams into the main application.
- **Game Design Lead – Logic Puzzles (Halil Köybaşı):**
  Designed and coded the 'Laser Mirrors' (Level 1) logic puzzle using Pygame, focusing on algorithmic pathfinding logic.
- **Game Design Lead – Physics Engine & Sensor Firmware (İsmail Efe Tekin):**
  Designed the 'Tilt Arena' (Level 2) mechanics and ball physics. Additionally, developed the firmware for the Sensor Unit (Arduino 2) to process MPU6050 accelerometer data and transmit it to the PC via Serial.
- **Game Design Lead – Resource Management (Fatih Kerim Yıldız):**
  Developed the 'Energy Stabilizer' (Level 3) minigame mechanics, balancing energy values for the user.
- **Game Design Lead – Computer Vision (Beyza Ekizoğlu):**
  Implemented the 'Red Light Green Light' (Level 4) using OpenCV for real-time motion detection via the camera.
- **Presentation & Media Production Team (Feyzanur Yazıcı & Büşra Uflaz):**
  Responsible for the project presentation and media assets. Created the PowerPoint slides, edited the final 5-minute Demo Video required for submission.

6. Challenges and Conclusion

The biggest challenge was the pin conflict mentioned in the hardware section. We learned that planning the pinout map before coding is essential. Also, synchronizing two different independent systems (Windows PC and Arduino) required precise timing in our code. Throughout this process, utilizing AI assistance proved to be a valuable asset, enhancing our problem-solving capabilities and helping us refine the system's architecture efficiently.

In conclusion, we successfully built a working prototype that is fun to play and demonstrates the interaction between high-level software and low-level hardware control.

Project Links

Demo Video: https://www.youtube.com/watch?v=pYHDOQSZFzM